CS 24 AP Computer Science A Review

Week 8: 2D Arrays

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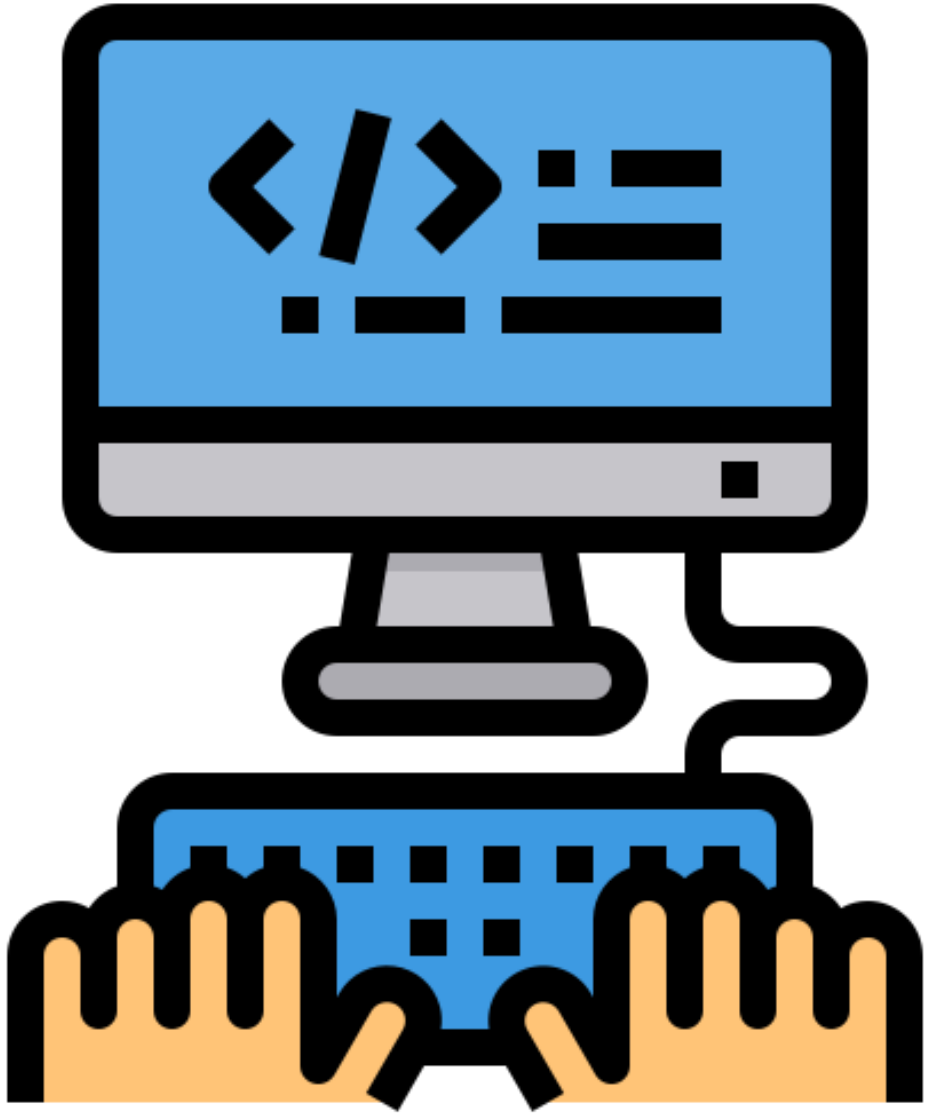
IEEE SENIOR MEMBER

Topics

- Array of Arrays
- Sum, Max, MaxIndex, Shuffle.
- Upper Triangle/Lower Triangle for K-Graph
- Row-Major Versus Column Major
- 1-D array conversion to 2-D conversion. 2-D Conversion to 1-D Conversion
- Mirror, Horizontal Flip, Vertical Flip



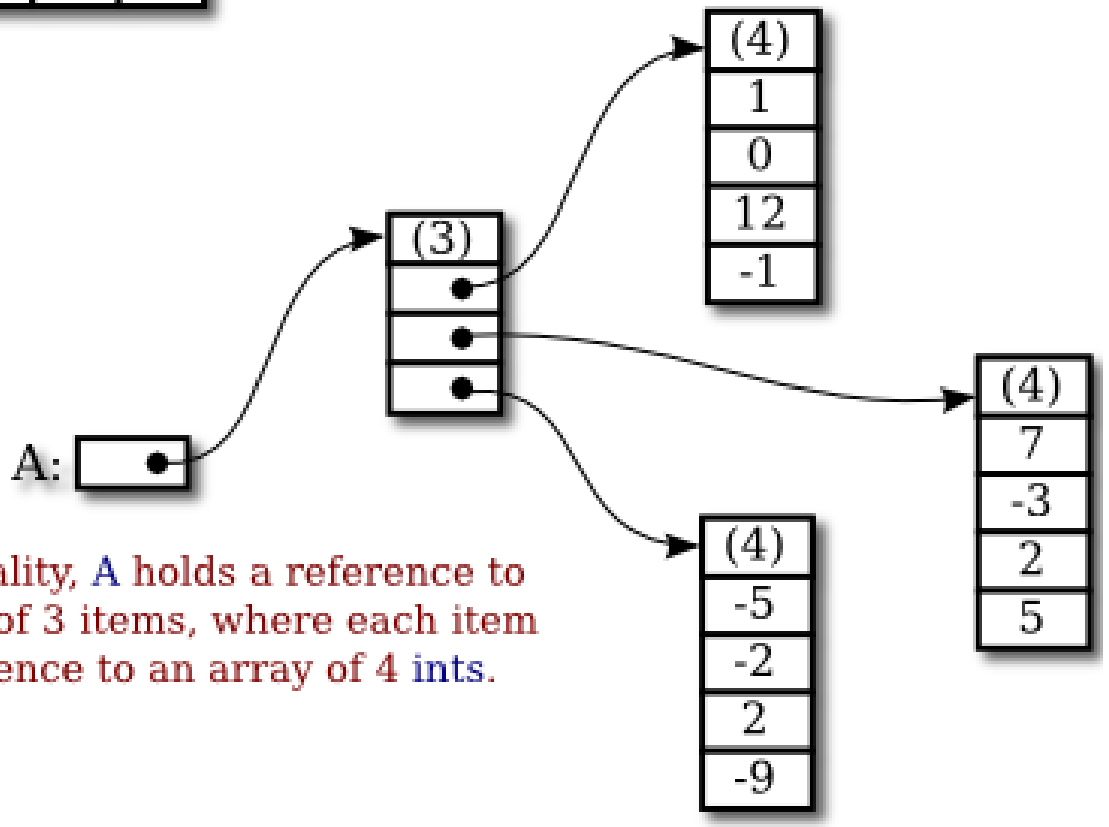
2D Arrays

SECTION 1

If you create an array A = new int[3][4], you should think of it as a "matrix" with 3 rows and 4 columns.

A:

1	0	12	-1
7	-3	2	5
-5	-2	2	-9



But in reality, A holds a reference to an array of 3 items, where each item is a reference to an array of 4 ints.

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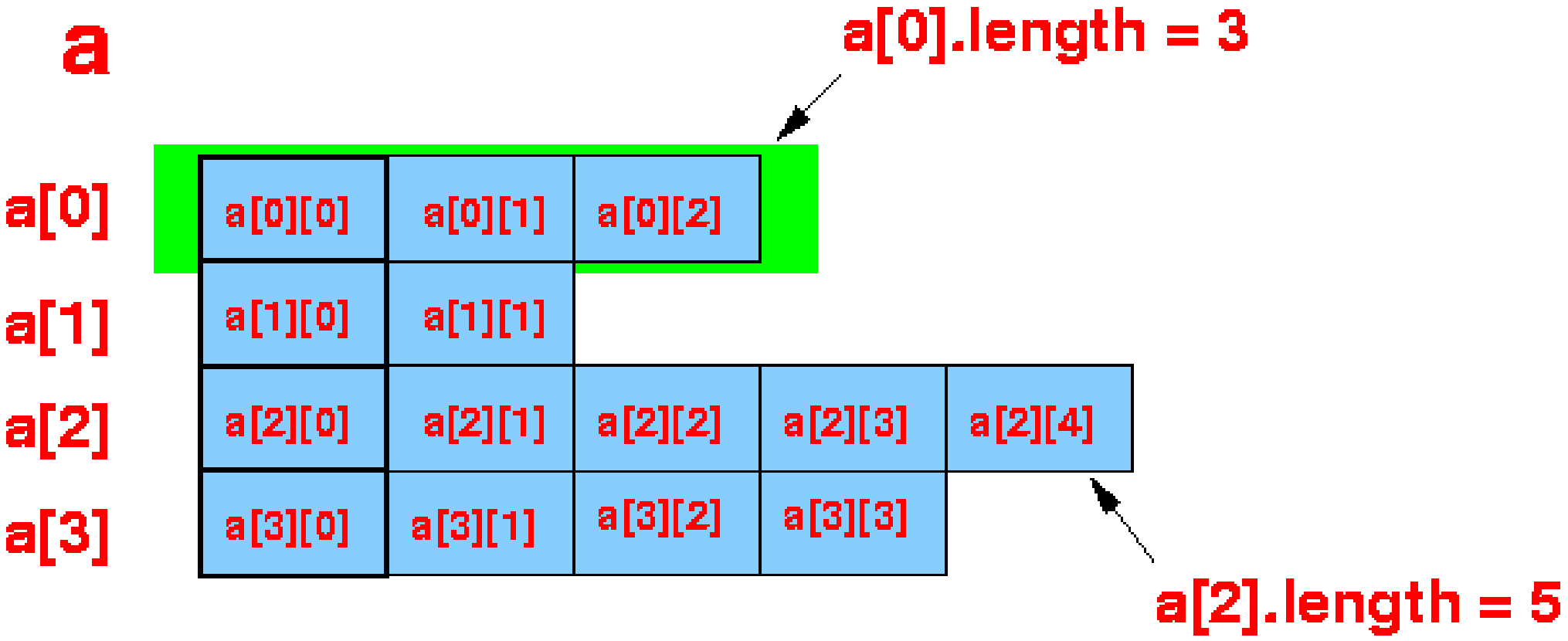
General Form for Creating 2-D Array Using a Pre-defined List of Data

```
datatype[][] nameOf2DArray = {
  {value1, value2, value3},
  {value4, value5, value6},
  { …, …, …}
};
```

```
String[][] candyBoard = {
 {“Jelly Bean”, “Lozenge”, “Lemon Drop”},
 {“Gum Square”, “Lollipop Head”, “Jujube Cluster”},
 {“Lozenge”, “Lollipop Head”, “Lemon Drop”},
 {“Jelly Bean”, “Lollipop Head”, “Lozenge”}
};
```


a
a[0].length = 3
a[0]
a[1]
a[2]
a[3]
a[0][0]
a[0][1]
a[0][2]
a[1][0]
a[1][1]
a[2][0]
a[2][1]
a[2][2]
a[2][3]
a[2][4]
a[3][0]
a[3][1]
a[3][2]
a[3][3]
a[2].length = 5

Using the length Field to Find the Number of Rows and Columns

Retrieve the number of rows from a 2-D array:

```
double[][] myBoard = new double[8][3];
int result = myBoard.length;
```

Retrieve the number of columns from a 2-D array:

```
double[][] myBoard = new double[8][3];
int result1 = myBoard[0].length;
int result2 = myBoard[5].length;
```

Multiple Choice (1)

Consider the following code segment:

```
int[][] num = new int[4][4];
for (int i=0; i<num.length; i++){
    for (int k=0; k<num.length; k++){
        num[i][k] = i * k;
    }
}
```

What are the contents of `num` after the code segment has executed?

Multiple Choice (1)

Ⓐ

0	0	0	0
0	1	2	3
0	2	4	6
0	3	6	9

Ⓑ

0	1	2	3
1	2	3	4
2	3	4	5
3	4	5	6

Ⓒ

1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4

Ⓓ

0	0	0	0
1	2	3	4
2	4	6	8
3	6	9	12

Multiple Choice (1)

Ⓐ

0	0	0	0
0	1	2	3
0	2	4	6
0	3	6	9

Ⓑ

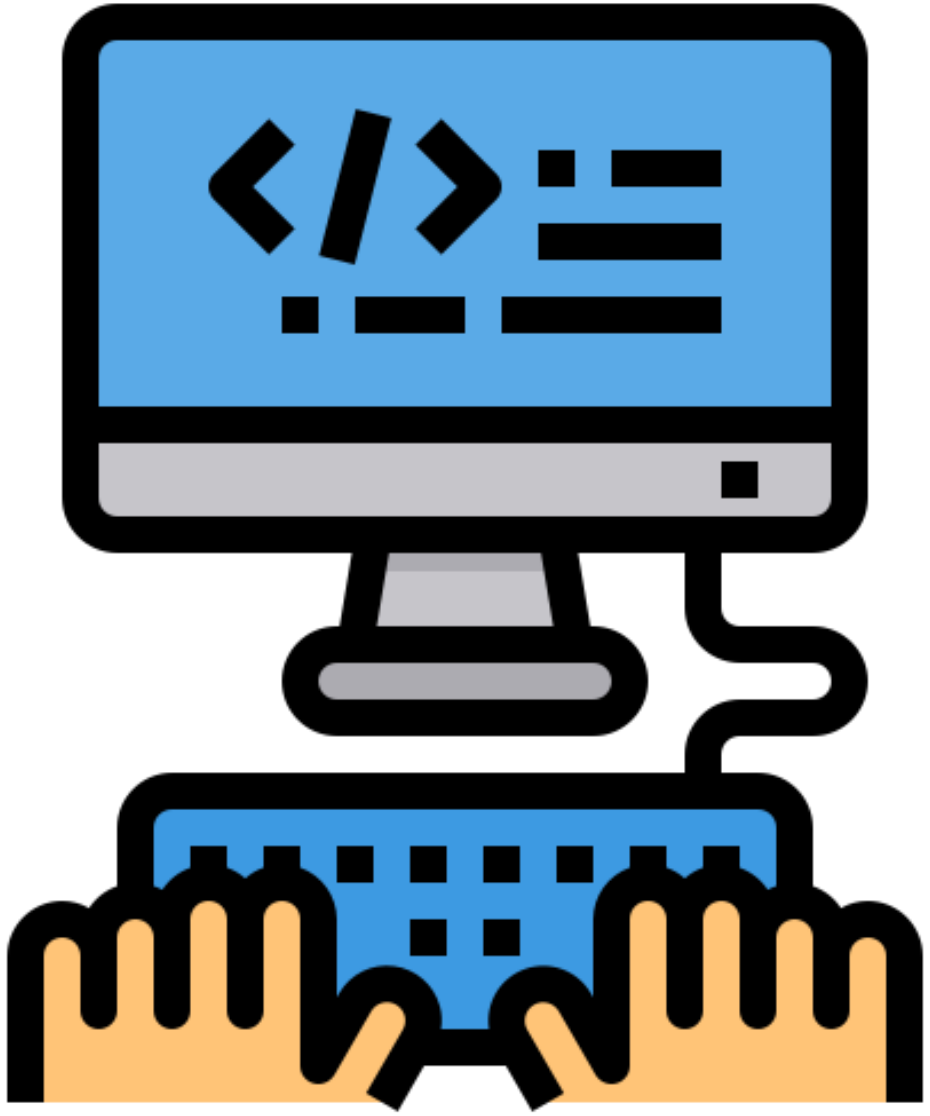
0	1	2	3
1	2	3	4
2	3	4	5
3	4	5	6

Ⓒ

1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4

Ⓓ

0	0	0	0
1	2	3	4
2	4	6	8
3	6	9	12

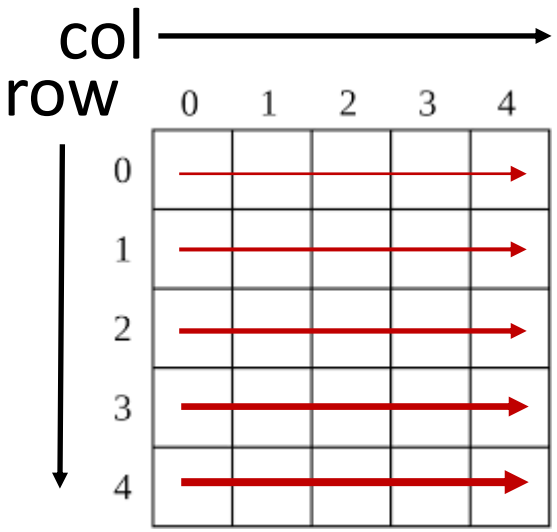


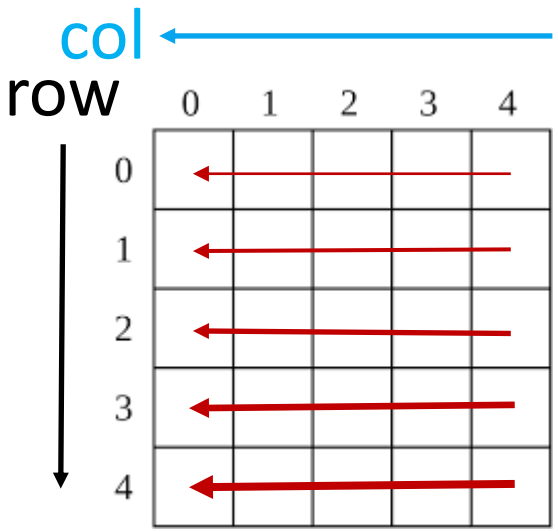
2D Traversal

SECTION 2

2-D Array Indexing for Traversal

int row, col; int[][] m=new int[5][5];

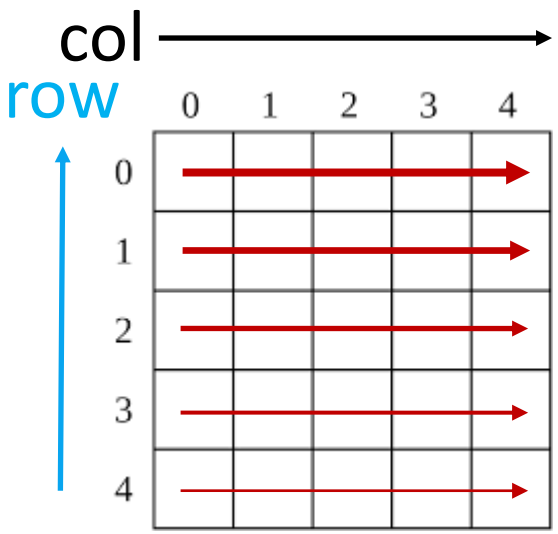


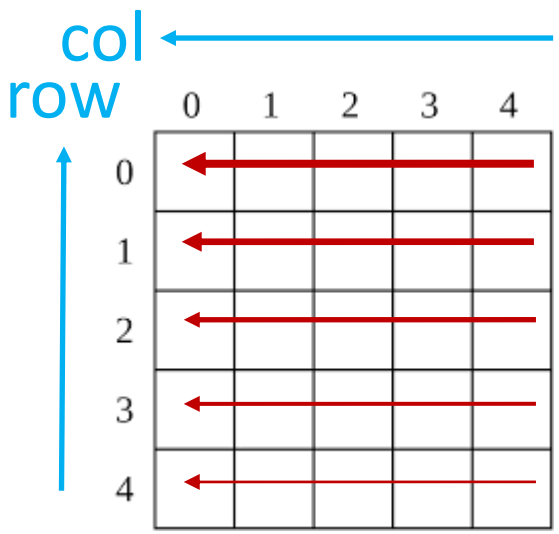



```
for(row=0; row<m.length; row++)
  for(col=0; col<m[0].length; col++)
      System.out.println(m[row][col]);
```

```
for(row=0; row<m.length; row++)
  for(col=m[0].length-1; col>=0; col--)
      System.out.println(m[row][col]);
```

2-D Array Indexing for Traversal
int row, col; int[][] m=new int[5][5];



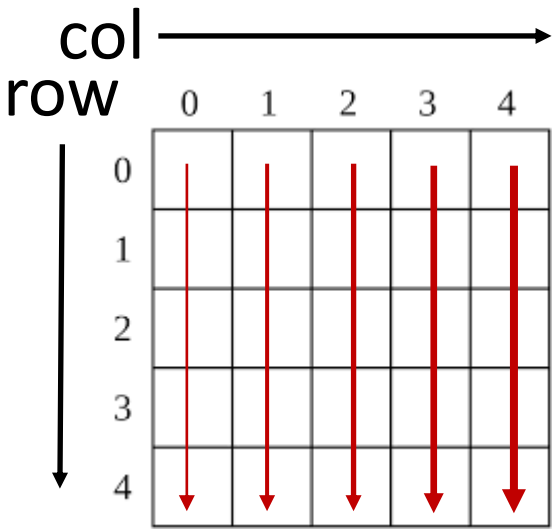


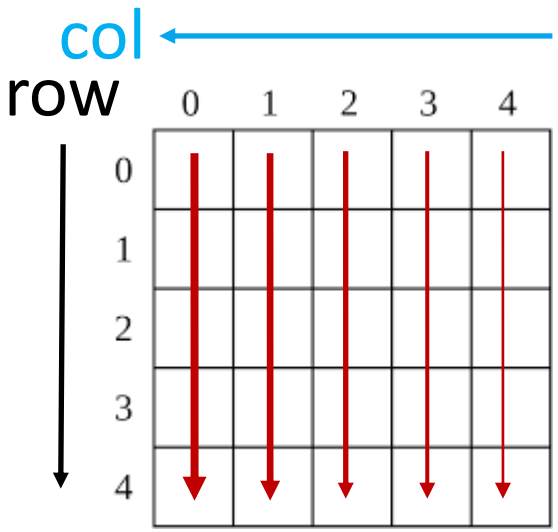

```
for(row=m.length-1; row>=0; row--)
  for(col=0; col<m[0].length; col++)
      System.out.println(m[row][col]);
```

```
for(row=m.length-1; row>=0; row--)
  for(col=m[0].length-1; col>=0; col--)
      System.out.println(m[row][col]);
```

2-D Array Indexing for Traversal

int row, col; int[][] m=new int[5][5];

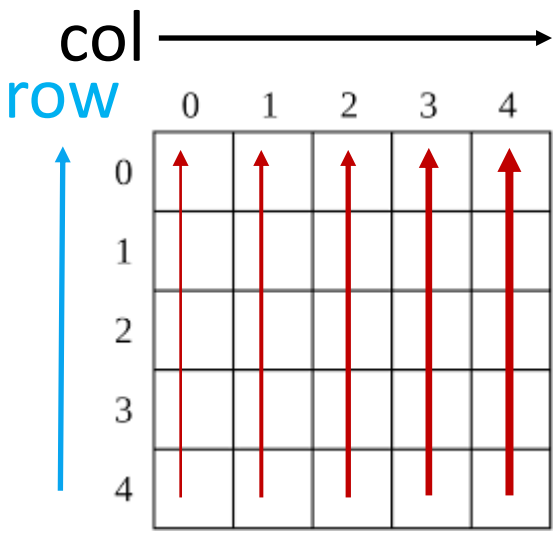


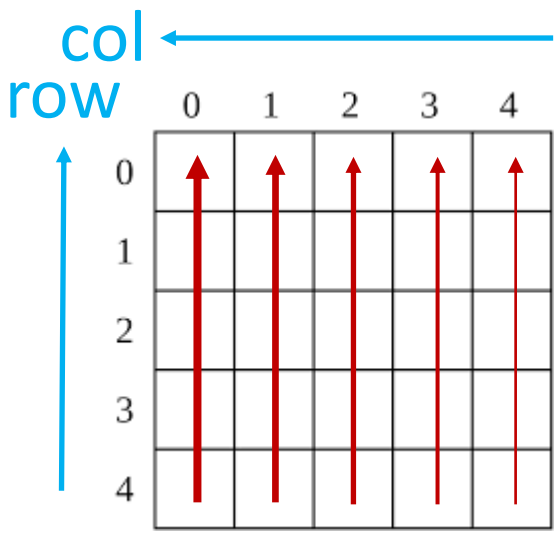



```
for(col=0; col<m[0].length; col++)
  for(row=0; row<m.length; row++)
      System.out.println(m[row][col]);
```

```
for(col=m[0].length-1; col>=0; col--)
  for(row=0; row<m.length; row++)
      System.out.println(m[row][col]);
```

2-D Array Indexing for Traversal
int row, col; int[][] m=new int[5][5];

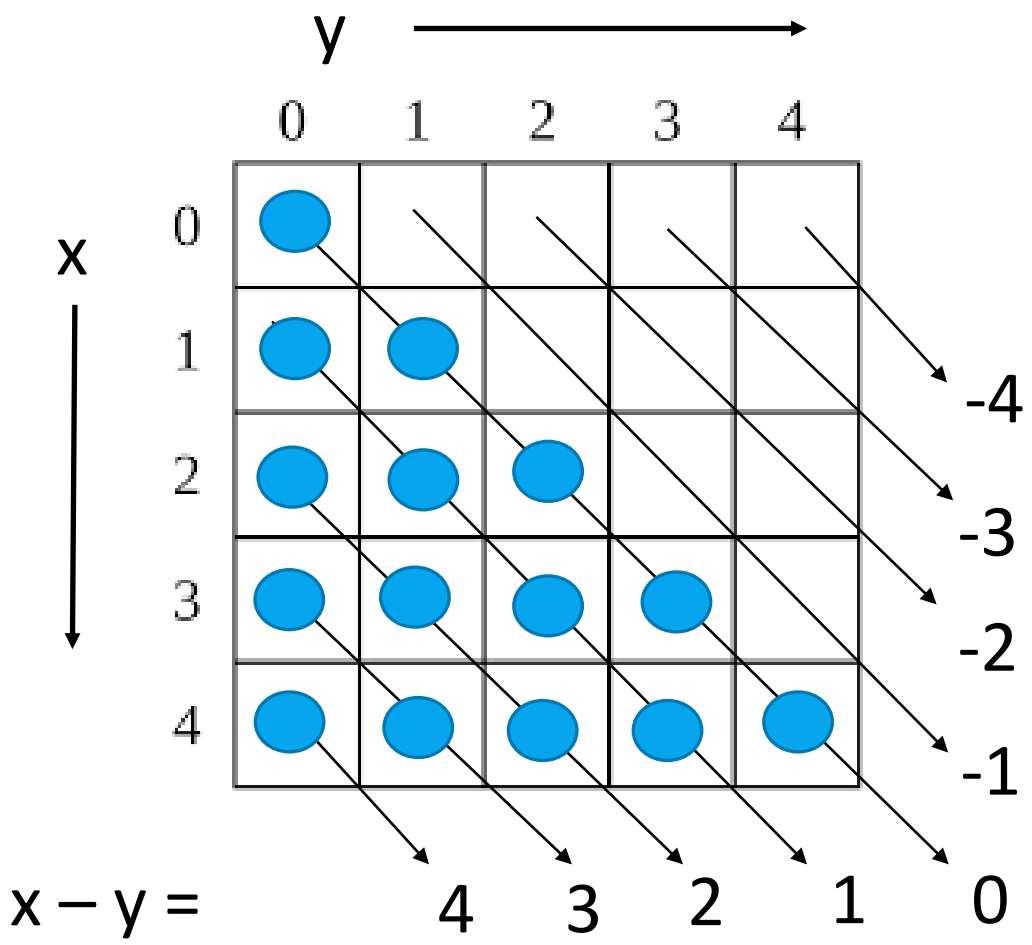





```
for(col=0; col<m[0].length; col++)
  for(row=m.length-1; row>=0; row--)
    System.out.println(m[row][col]);
```

```
for(col=m[0].length-1; col>=0; col--)
  for(row=m.length-1; row>=0; row--)
    System.out.println(m[row][col]);
```

Partial Array Traversal



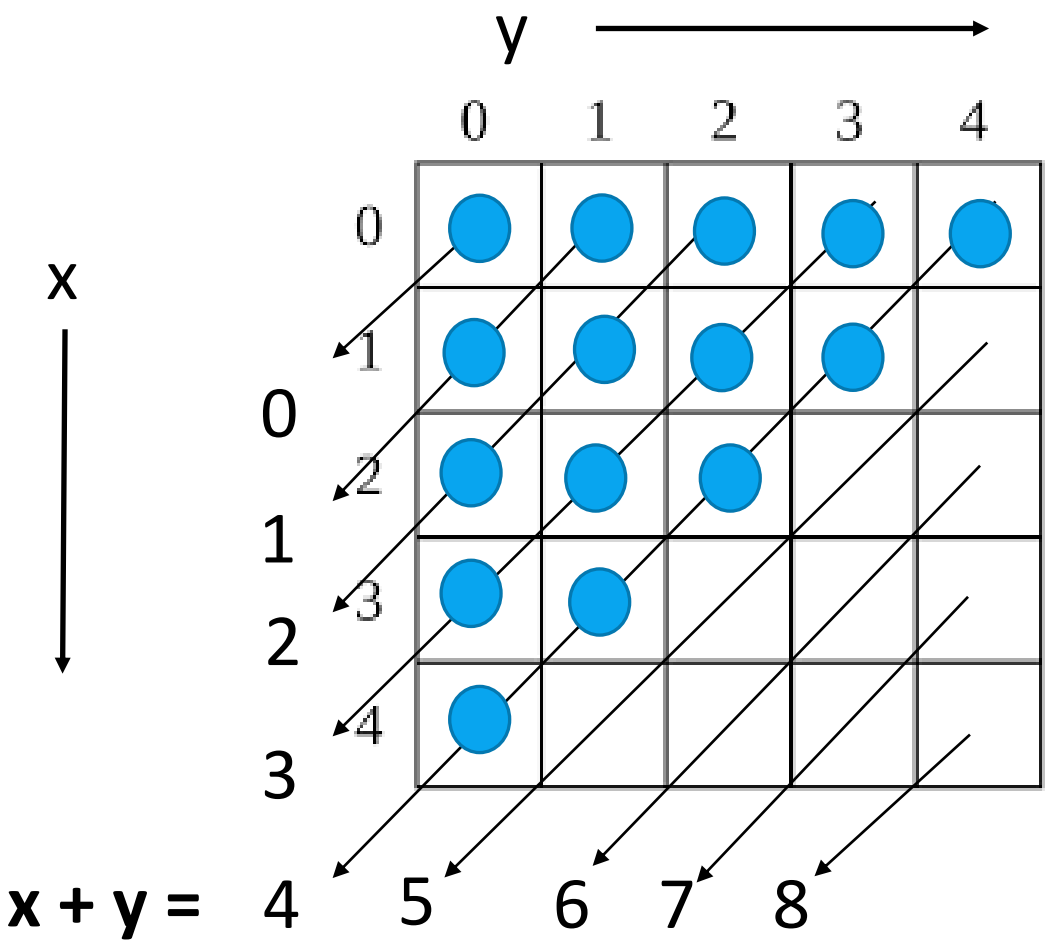

```
for (int I = 0; i<m.length; i++)
   for (int j =0; j< i +1;  j ++)
      { /* do something */}
```

Index:

Stop Condition: j stop at j = i.

i – j = 0;

Partial Array Traversal



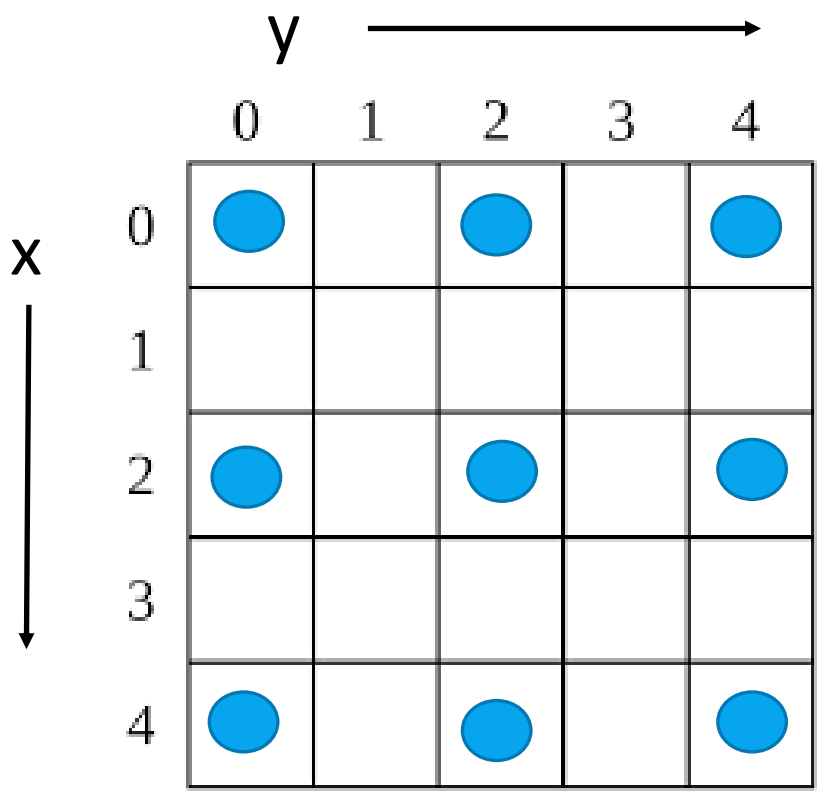

```
for (int i = 0; i<m.length; i++)
   for (int j =m.length-1-i; j>=0;  j --)
      { /* do something */}
```

Index:

Start Condition: i + j = m.length-1;

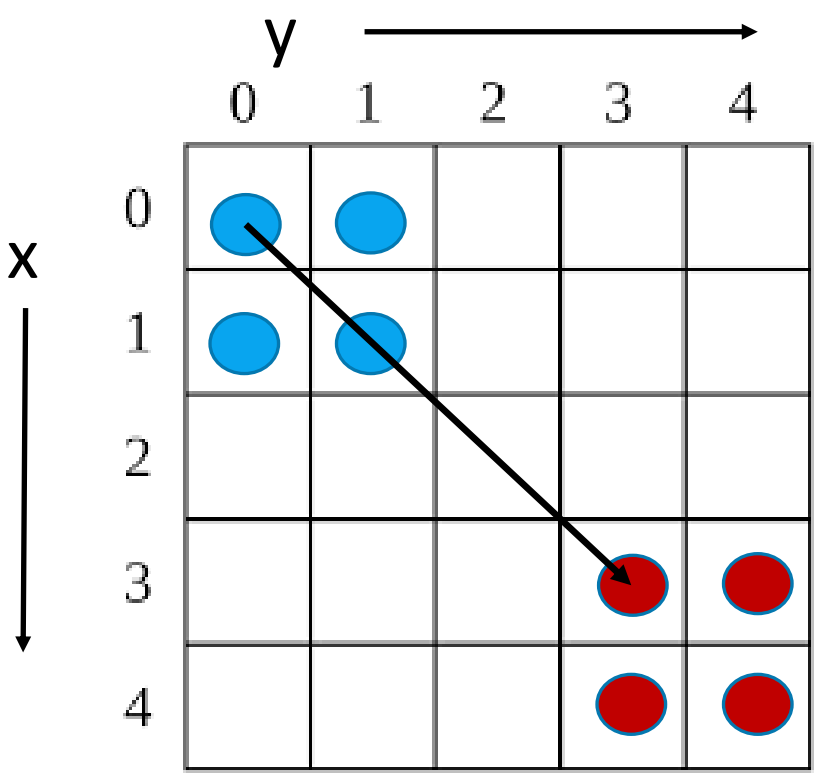
i + j = 4;

Partial Array Traversal

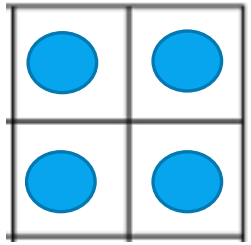



```
for (int i = 0; i<m.length; i+=2)
   for (int j =0; j<m[0].length;  j+=2)
      { /* do something */}
```

Vector Operation (Area Copy)




```
for (int i = 0; i<2; i++)
  for (int j=0; j<2; j++)
    m[3+i][3+j] = m[0+i][0+j];
      /* 0 is not needed */
```



Area to be Copied: **2 x 2** block

From (0, 0) to (3, 3)

If there is overlap, we need to put the area into a buffer first.

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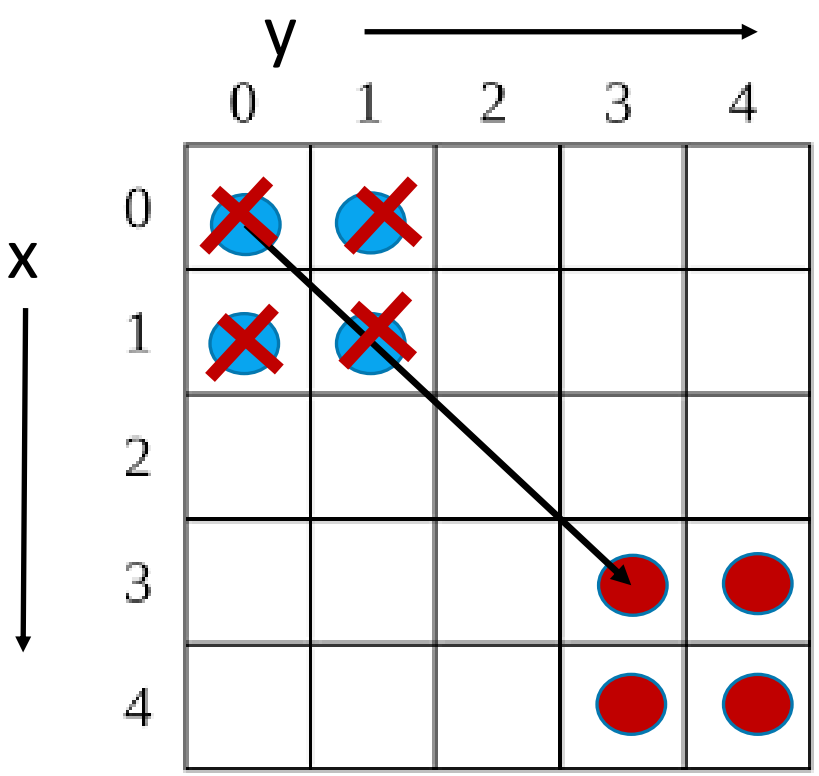
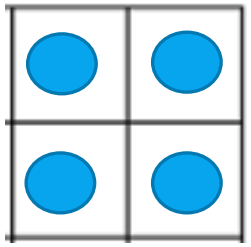
Vector Operation (Area Move)



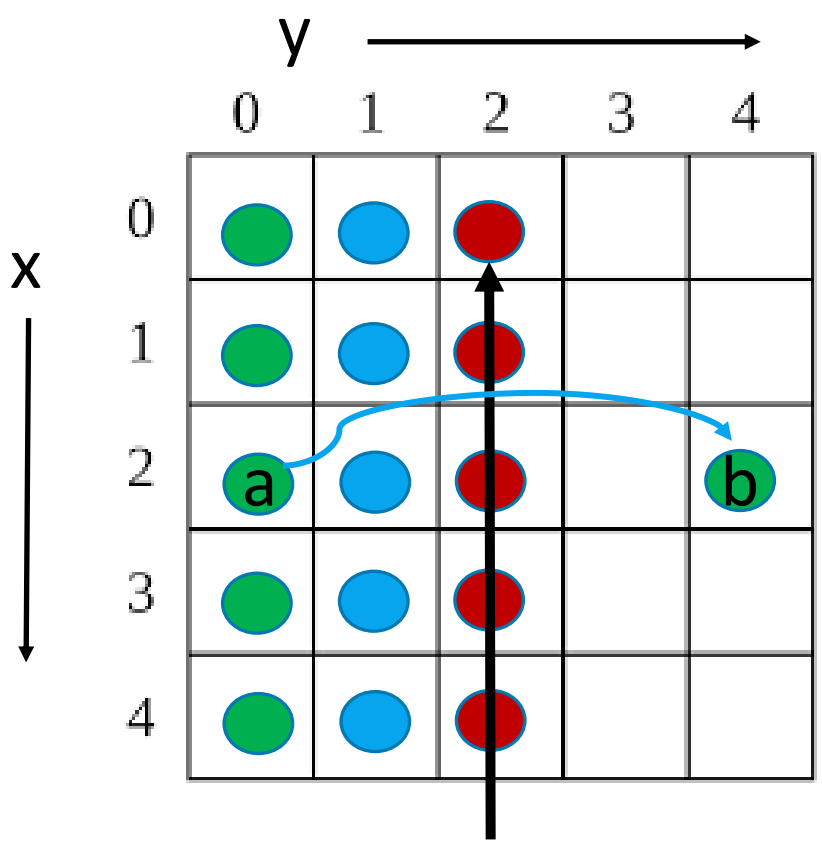

```
for (int i = 0; i<2; i++)
   for (int j=0; j<2; j++)
      m[3+i][3+j] = m[0+i][0+j];
for (int i = 0; i<2; i++)
   for (int j=0; j<2; j++)
      m[0+i][0+j]=0;
```



Area to be Copied: **2 x 2** block

From (0, 0) to (3, 3)

Flip



Symmetric Line for Flipping:

j = 2; // j = m.length/2

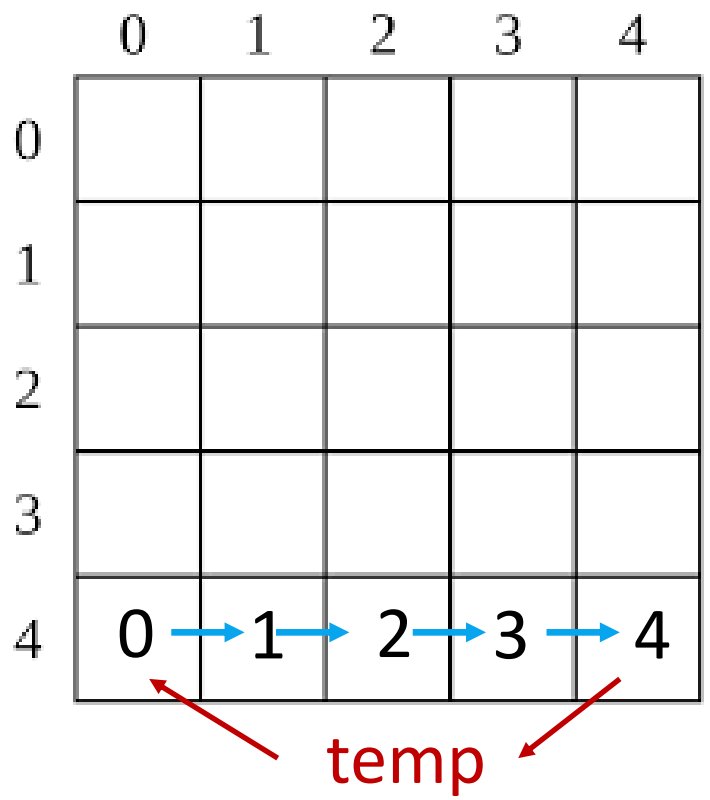
(a + b)/2 = 2;

b == 4-a; // b = m.length –a-1;

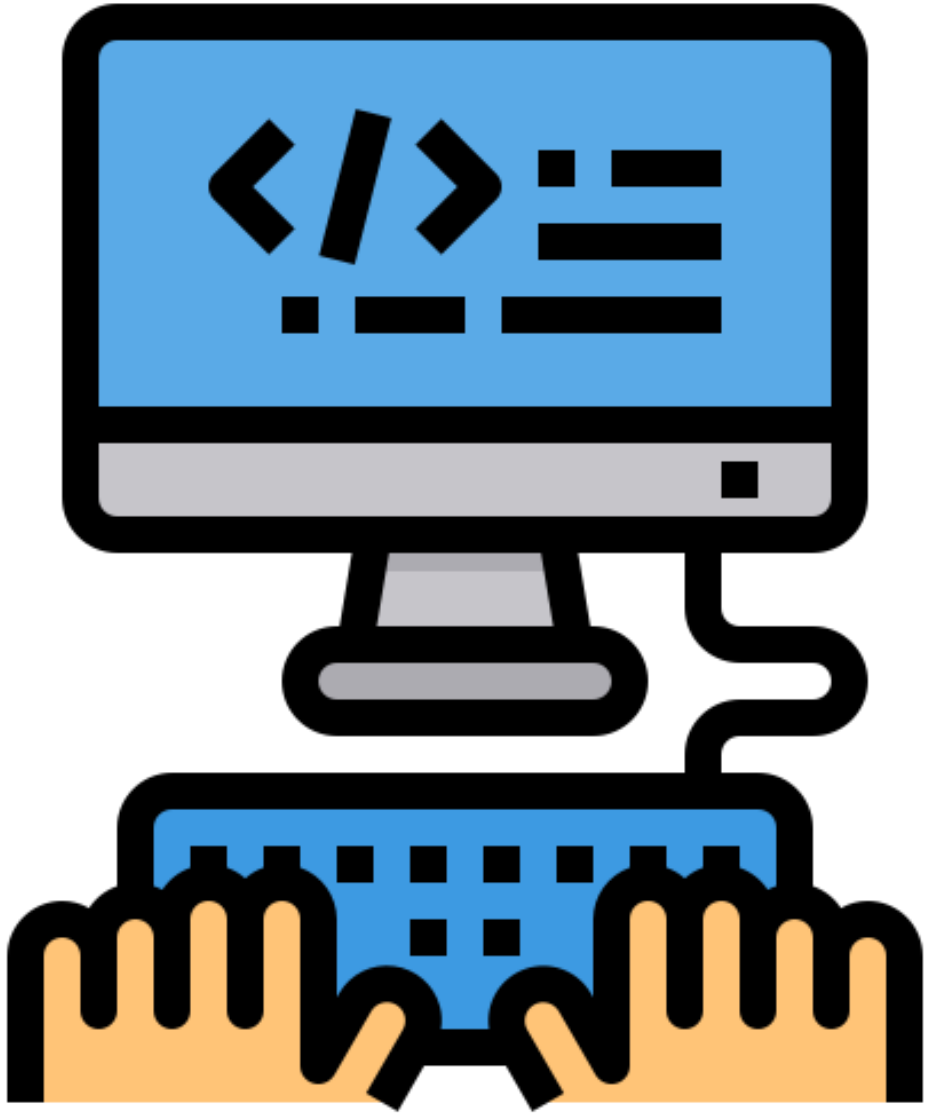
```
for (int i=0; i<m.length; i++){
   for (int j=0; j<m.length/2; j++){
      m[i][m.length-j-1] = m[i][j];
  }
}
```

Area Shift

```
for (int i=0; i<m.length; i++){
    int temp = m[i][m.length-1];
    for (int j=m[0].length-2; j<=0; j--){
        m[i][j+1] = m[i][j];
    }
    m[i][0] =temp;
}
```

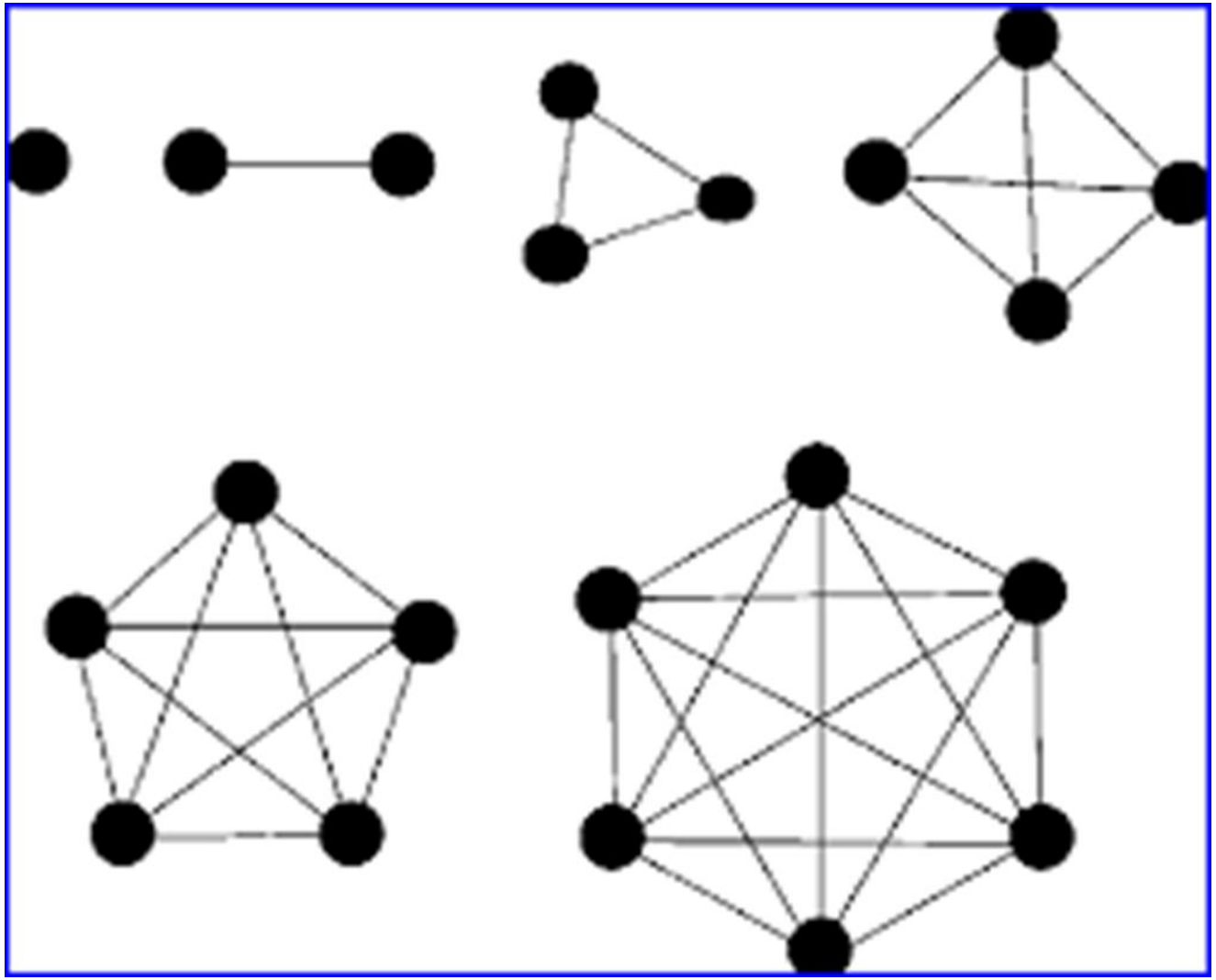


One Row after Shift: 4 0 1 2 3



2D Array Processing

SECTION 3



K-Graph Problems

Combination Problem

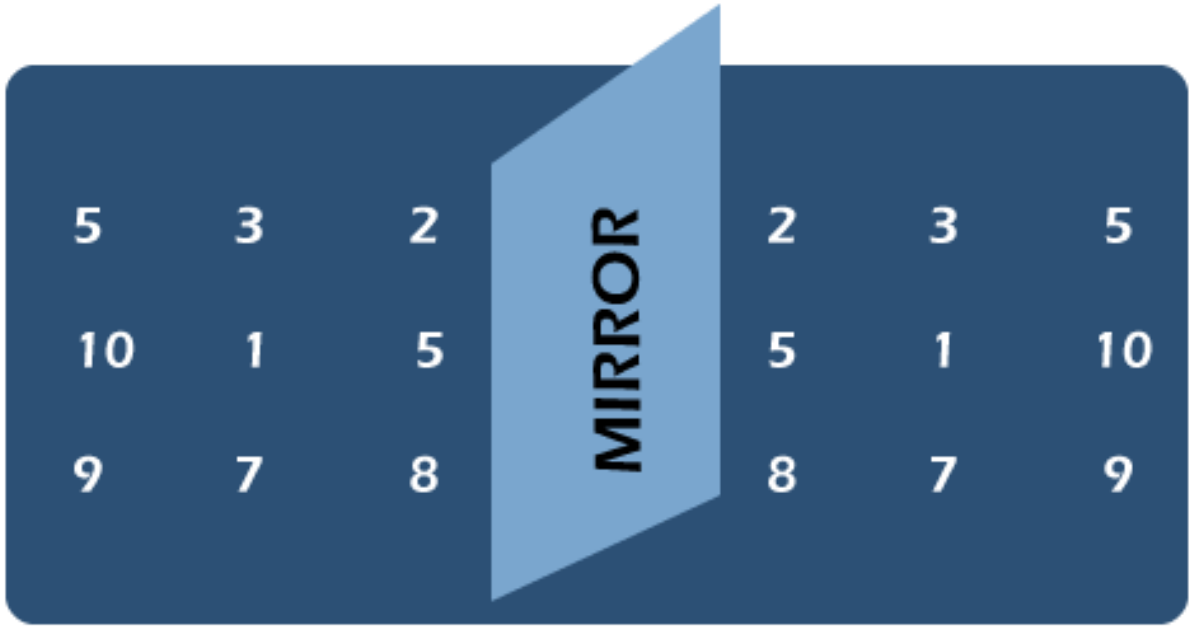
```
static String s = "ABCDE";

public static void combination(String s){
   for (int i=0; i<s.length()-1; i++){
        for (int j=i+1; j<s.length(); j++){
            String x = ""+s.charAt(i)+s.charAt(j);
            System.out.println(x);
         }
     }
}
```

Transpose

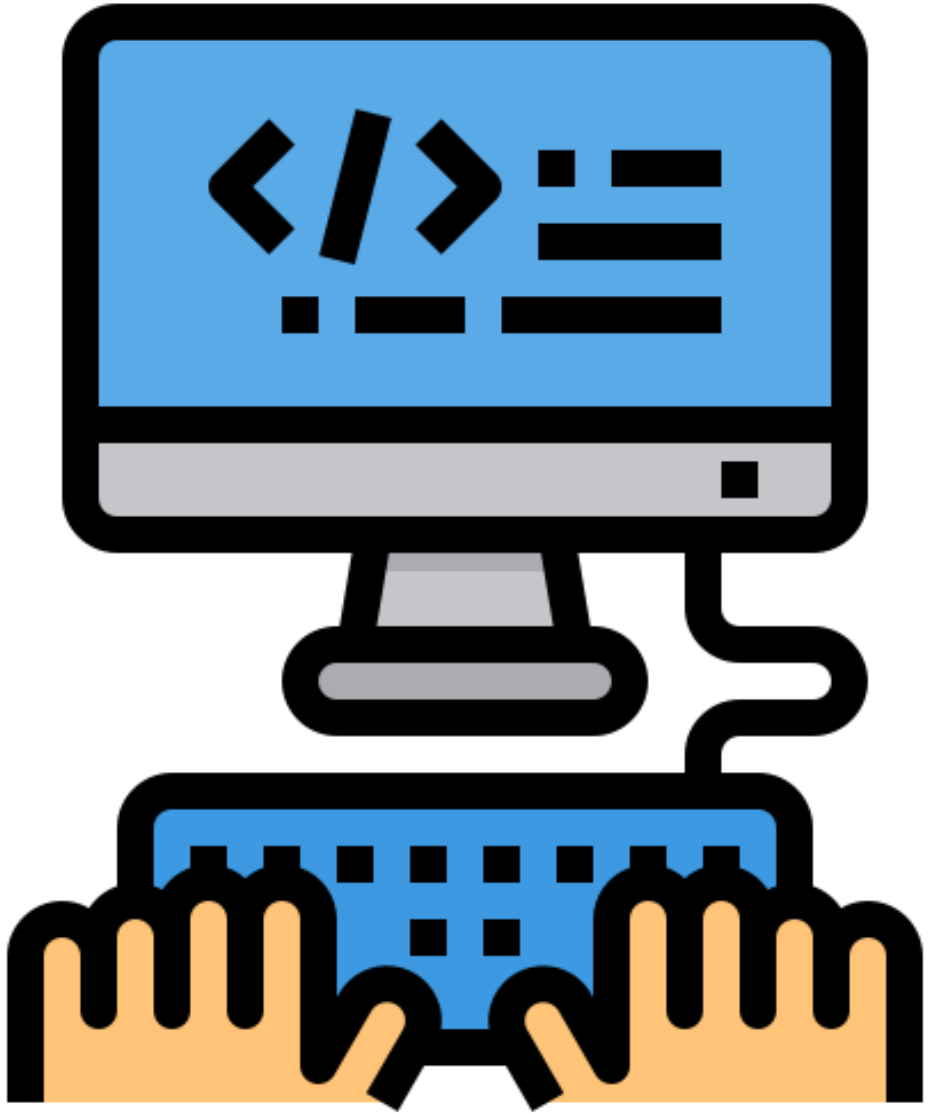
2	4	-1
-10	5	11
18	-7	6

2	-10	18
4	5	-7
-1	11	6

Mirror



Mirror Image of a 2D Array



Practice MCQ Questions

SECTION 5

Multiple Choice (1)

The following class `Arr2d`, is meant to represent a two-dimensional array object. The constructor will initialize `Arr2d` using the number of rows and columns that have been passed. Choose the statement that will initialize the array in the constructor.

```
Public class Arr2D{
  private int[][] arr;
  Arr2D (int rows, int columns){
     /* missing code */
  }
}
```

Multiple Choice (1)

Ⓐ `int[] arr = new int[rows][columns];`

Ⓑ `int[][] arr = new int[rows-1][columns-1];`

Ⓒ `arr = new int[rows][columns];`

Ⓓ `arr = new int[rows-1][columns-1];`

Multiple Choice (1)

Ⓐ `int[] arr = new int[rows][columns];`

Ⓑ `int[][] arr = new int[rows-1][columns-1];`

Ⓒ `arr = new int[rows][columns];`

Ⓓ `arr = new int[rows-1][columns-1];`

Multiple Choice (2)

Consider the following code segment:

```
int[][] numbers = {{1, 2, 3, 4},
                   {5, 6, 7, 8}};
for (int[] nums: numbers)
    for (int n: nums)
        System.out.print(n + " ");
System.out.print("\n");
```

What is printed das a result of executing the code segment?

Multiple Choice (2)

Ⓐ 1 2 3 4 5 6 7 8

Ⓑ 1 2 3 4
5 6 7 8

Ⓒ 1 2
3 4
5 6
7 8

Ⓓ A Compiler error would occur.

Multiple Choice (2)

Ⓐ 1 2 3 4 5 6 7 8

Ⓑ 1 2 3 4

5 6 7 8

Ⓒ 1 2

3 4

5 6

7 8

Ⓓ A Compiler error would occur.

Multiple Choice (3)

Consider the following code segment:

```
int[][] numbers = new int[4][4];
initializedIt(numbers);
int total =0;
for (int z=0; z<numbers.length; z++){
  total += numbers[z][numbers[0].length-1-z];
}
```

Multiple Choice (3)

The call to `initializedIt()` on the second line initializes the array numbers so that it looks like the following:

```
1 2 5 3
7 9 4 0
3 3 2 5
4 5 8 1
```

What will be the value of `total` after the code has executed?

Multiple Choice (3)

Ⓐ 11

Ⓑ 12

Ⓒ 13

Ⓓ 14

Multiple Choice (3)

Ⓐ 11

Ⓑ 12

Ⓒ 13

Ⓓ 14

Multiple Choice (4)

Consider the following code segment:

```
int[][] numbers = new int[3][6];
initializedIt(numbers);
int total =0;
for (int j=0; j<numbers.length; j++){
  for (int k=0; k<numbers[0].length; k+=2){
     total += numbers[j][k];
  }
}
```

Multiple Choice (4)

The call to `initializedIt()` on the second line initializes the array `numbers` so that it looks like the following:

```
2 4 6 3 2 1
5 6 7 4 2 9
4 0 5 6 4 2
```

What will be the value of `total` after the code has executed?

Multiple Choice (4)

Ⓐ 18

Ⓑ 35

Ⓒ 37

Ⓓ 72

Multiple Choice (4)

Ⓐ 18

Ⓑ 35

Ⓒ 37

Ⓓ 72

Multiple Choice (5)

Consider the following classes:

```
int[][] num = new int[4][4];
for (int i=0; i<num.length; i++){
  for (int k=0; k<num[0].length; k++){
     num[i][k] = i * k;
  }
}
```

What are the content of `num` after the code segment has executed?

Multiple Choice (5)

Ⓐ

0	0	0	0
0	1	2	3
0	2	4	6
0	3	6	9

Ⓑ

0	1	2	3
1	2	3	4
2	3	4	5
3	4	5	6

Ⓒ

1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4

Ⓓ

0	0	0	0
1	2	3	4
2	4	6	8
3	6	9	12

Multiple Choice (5)

Ⓐ

0	0	0	0
0	1	2	3
0	2	4	6
0	3	6	9

Ⓑ

0	1	2	3
1	2	3	4
2	3	4	5
3	4	5	6

Ⓒ

1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4

Ⓓ

0	0	0	0
1	2	3	4
2	4	6	8
3	6	9	12

Multiple Choice (6)

Consider the following code segment:

```
int[][] numbers = {{1, 2, 3, 4}, {5, 6, 7, 8}};
initializedIt(numbers);
String line = "";
for (int a =0; a<numbers[0].length; a++){
  for (int b=0; b<numbers.length; b++){
      line += numbers[b][a]+ " ";
   }
   line += "\n";
}
System.out.println(line);
```

Multiple Choice (6)

The call to initializedIt() on the second line initializes the array so that it looks like the following:

```
1 2 3 4
5 6 7 8
```

What will be printed after the code has executed?

Multiple Choice (6)

Ⓐ

```
1 2 3 4
5 6 7 8
```

Ⓑ

```
5 6 7 8
1 2 3 4
```

Ⓒ

```
8 7 6 5
4 3 2 1
```

Ⓓ

```
1 5
2 6
3 4
4 8
```

Multiple Choice (6)

Ⓐ

1	2	3	4
5	6	7	8

Ⓑ

5	6	7	8
1	2	3	4

Ⓒ

8	7	6	5
4	3	2	1

Ⓓ

1	5
2	6
3	4
4	8

Multiple Choice (7)

Consider the following client code and method:

```
int[][] numbers = {{1, 2, 3, 4},
        {5, 6, 7, 8}, {9, 10, 11, 12}};
int [] myArr = mystery(numbers, 2);
for (int j: myArr) System.out.print(j+" ");

public static int[] mystery(int[][] arr, int n){
    int[] m = new int[arr.length];
    for (int i=0; i<arr.length; i++)
      m[i] = arr[i][n];
    return m;
}
```

Multiple Choice (7)

What will be printed after the code executed?

Ⓐ `2 6 10`

Ⓑ `9 10 11 12`

Ⓒ `4 8 12`

Ⓓ `3 7 11`

Multiple Choice (7)

What will be printed after the code executed?

Ⓐ `2 6 10`

Ⓑ `9 10 11 12`

Ⓒ `4 8 12`

Ⓓ `3 7 11`

Multiple Choice (8)

Consider the following code:

```
int[][] numbers = {{1, 2, 3, 4},
        {5, 6, 7, 8}, {9, 10, 11, 12}};
for (int r = numbers.length-1; r>0; r--){
  for (int c=0; c<numbers[r].length; c++){
     numbers[r-1][c] = numbers[r][c];
  }
}
```

Which of the following represents the current values in the array?

Multiple Choice (6)

Ⓐ

1	2	3	4
5	6	7	8
9	10	11	12

Ⓑ

9	10	11	12
5	6	7	8
1	2	3	4

Ⓒ

1	2	3	4
1	2	3	4
1	2	3	4

Ⓓ

9	10	11	12
9	10	11	12
9	10	11	12

Multiple Choice (6)

Ⓐ

1	2	3	4
5	6	7	8
9	10	11	12

Ⓑ

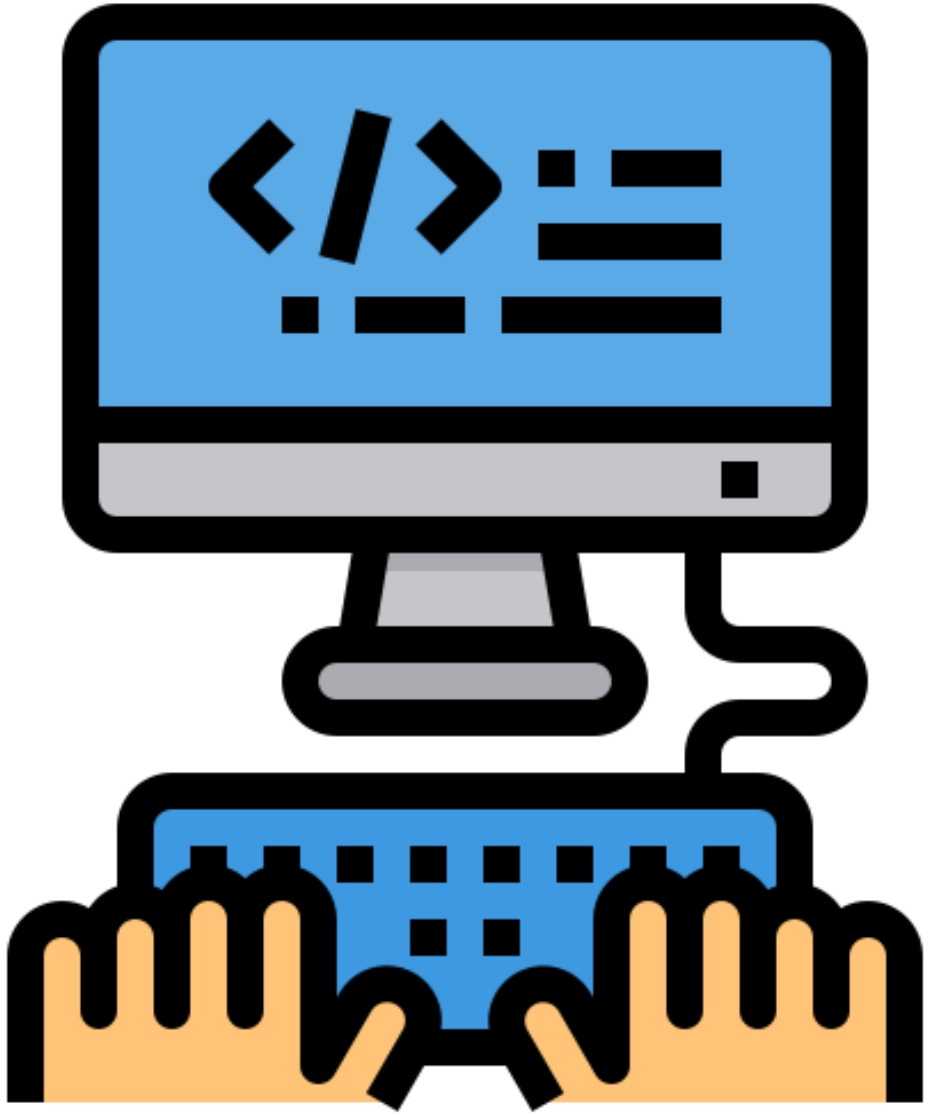
9	10	11	12
5	6	7	8
1	2	3	4

Ⓒ

1	2	3	4
1	2	3	4
1	2	3	4

Ⓓ

9	10	11	12
9	10	11	12
9	10	11	12



2D Array Algorithms

SECTION 6

Zig-Zag Scan

```
public void zigzag(int[][] m){
    for (int r=0; r<m.length; r++){
      if (i%2==0){
        for (int c=0; c<m[r].length; c++)
          System.out.print(m[r][c]+ " ");
      }
      else{
        for (int c=m[r].length-1; c>=0; c--)
          System.out.print(m[r][c]+ " ");
      }
      System.out.println();
    }
}
```

Copy 1-D Array to 2-D Array

```
public  int[][] copy(int[] a, int[][] m){
    int p=0;
    for (int r=0; r<m.length; r++){
        for (int c=0; c<m[r].length; c++){
          if (p<a.length) m[r][c] = a[p++];
        }
    }
    return m;
}
```

Copy 2-D Array to 1-D Array

```
public  int[] copy(int[] a, int[][] m){
    int p=0;
    for (int r=0; r<m.length; r++){
        for (int c=0; c<m[r].length; c++){
          if (p<a.length) a[p++] = m[r][c];
        }
    }
    return a;
}
```